Venkata Lakshmi Pasupuleti

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PROFESSIONAL SUMMARY

AI Data Engineer with 8+ years building **conversational AI agents**, data engineering pipelines, and production systems at scale. Built **Leasel** - an 8-agent LangGraph platform for **customer contact analysis, transcript summarization, and root-cause investigation**: classifies support tickets, extracts ranked evidence-backed hypotheses, produces **confidence-scored summaries with audit trails**, and routes low-confidence cases to human escalation. Built **AgentProbe** - validation mechanisms verifying AI output accuracy against source data (CI-integrated 5-metric eval, blocks failing runs automatically). At Entrata (PropTech, 100K+ MAU), delivered **ETL pipelines, data modeling, and AWS data engineering** (S3, Glue, RDS) across 20+ production modules. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Conversational AI Agents | Contact Analysis & Summarization | Confidence Scoring & Audit Trails | Validation Mechanisms | Data Engineering & ETL | SQL & Data Modeling | RAG Pipelines | Prototype to Production

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - delivering a risk-aware compliance strategy: compliant missions export, non-compliant route to rejection with grounded reasons and audit trail.
- Implemented a **RAG retrieval layer** over FAA Part 107 corpus - every compliance check cited to source regulation; confidence-scored output with documented ChromaDB/embedding upgrade path.
- Engineered **validation guardrails** enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export; improved agent accuracy through iterative prompt engineering.

George Mason University - College of Engineering
Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Delivered technical workshops to **600+ students** across MIS 303 - Python programming, SQL, database design, and data analysis; weekly office hours, lab instruction, and rubric-aligned grading.

Entrata - PropTech / Real Estate SaaS
Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- **Shipped production code across 20+ full-stack modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime over 6+ consecutive months.
- **Built ETL pipelines and data engineering workflows** on AWS (S3, Glue, RDS) - processed 1.6M HUD/Tax Credit income limit records across 5 years; designed data models, ingestion pipelines, and schema architecture for 20+ production modules.
- **Shipped first production AI pipeline** on 1.6M financial records (Python, PyTorch, OpenAI API) - automated data processing replacing manual workflows; output validation gates, quality controls, and compliance guardrails at launch.
- **Optimized 200+ SQL queries** (30% performance gain) through execution plan tuning, indexing, database partitioning, and caching; led data modeling and database architecture decisions in collaboration with data architects.
- **Led cloud migration** to AWS S3 with Libsodium-encrypted pipelines - 100% PHI compliance, zero security incidents; converted 20% of modules to service-based microservices architecture (50% scalability improvement).
- Eliminated 80% of manual operational workload via Python/Bash automation; mentored 5+ junior engineers across 300+ peer code reviews.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Built full-stack property management modules (PHP, Python, PostgreSQL); ETL pipelines for 1.6M HUD/Tax Credit records; cloud migration to AWS S3; redesigned document storage database schema.

PROJECTS

Leasel: AI Contact Analysis & Support Investigation Platform 2026

Built an 8-agent LangGraph conversational AI platform for **customer contact analysis, transcript summarization, and root-cause investigation**. Support Investigation pipeline: Triage → Retrieval → Configuration → Historical Incident → Reasoning → Validation → Human Escalation - classifies incoming support tickets, extracts ranked evidence-backed root-cause hypotheses from business rules and historical incidents, produces **confidence-scored summaries with full audit trails**, and routes low-confidence cases to HITL escalation. Core design rule enforced in code: LLM extracts and reasons; deterministic logic or human makes every final regulated call. Achieved **93.6% precision** on 1,240 lease PDFs across 5 US states.

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite

AgentProbe: Agentic AI Testing & Validation Toolkit Live on GitHub

Built **validation mechanisms verifying AI output accuracy against source data** - CI-integrated 5-metric LLM-as-judge evaluation pipeline (factual accuracy, hallucination detection, retrieval quality, completeness, compliance); every commit scored automatically, regressions blocked before merge. Deployed 15-endpoint FastAPI backend with a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) and multi-provider routing (Anthropic, OpenAI, Groq, Ollama). React dashboard delivers trace-level observability: per-run metrics, regression drift, latency.

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI API · Anthropic API · React · Docker · GitHub Actions

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

AI & Agents: LangChain, LangGraph, Conversational AI Agents, Multi-Agent Orchestration, Tool Calling, ChromaDB, RAG Pipelines, Semantic Search, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Confidence Scoring, Validation Mechanisms, HITL, Guardrails, Prompt Engineering, Output Validation, Audit Trails, OpenAI API, Anthropic API, FastAPI

Data Engineering: ETL Pipelines, Data Modeling, Data Pipelines, AWS Glue, AWS S3, AWS RDS, Redshift (evaluating), PostgreSQL, MySQL, Oracle, ChromaDB (vector/non-relational), NoSQL, Ingestion Pipelines, Schema Design, SQL Optimization

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, Distributed Systems, RabbitMQ, Git, Linux, Agile/Scrum

ML & Data: PyTorch, Scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, NumPy, Pandas, spaCy, BERT, Tableau, Power BI